

The Complete Integer-Kink Dirac Atlas: Zero Mode, Bound Ladder, Thresholds, and Reflectionlessness

HCS-C352 source-local reconstruction

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Abstract

For every positive integer height, we give a single proof of the complete spectral and scattering atlas of the one-dimensional hyperbolic-tangent Dirac kink. Bounded-perturbation theory fixes the self-adjoint domain; supersymmetric factorization and shape invariance exhaust the simple bound ladder; the unpaired kernel gives exactly one chiral zero mode; and Darboux images of the free zero-momentum state and plane waves settle both threshold resonances and reflectionlessness. Exact ledgers independently check 25 heights and 150 rational complex scattering values, but the all-integer conclusion comes from the analytic descent argument. No priority or arithmetic-target claim is made.

Revision certificate. Domain, factorization, and complete discrete-ladder closure.

1 Frozen operator and theorem

On $L^2(\mathbb{R}; \mathbb{C}^2)$ define, for $n \in \mathbb{Z}_{\geq 1}$,

$$A_n = \frac{d}{dx} + n \tanh x, \quad A_n^* = -\frac{d}{dx} + n \tanh x, \quad H_n = \begin{pmatrix} 0 & A_n^* \\ A_n & 0 \end{pmatrix}, \quad (1)$$

with domain $H^1(\mathbb{R}; \mathbb{C}^2)$. “Upper” and “lower” always refer to the first and second spinor components in (1). A threshold resonance means a nonzero bounded distributional solution which is not in L^2 .

Theorem 1 (Integer-kink atlas). *For every integer $n \geq 1$ the following statements hold.*

- (i) H_n is self-adjoint on $H^1(\mathbb{R}; \mathbb{C}^2)$ and generates the unitary group e^{-itH_n} .
- (ii) Its essential spectrum is

$$\sigma_{\text{ess}}(H_n) = (-\infty, -n] \cup [n, \infty),$$

and the continuous spectral subspace is purely absolutely continuous.

- (iii) The complete point spectrum is

$$\sigma_{\text{p}}(H_n) = \{0\} \cup \left\{ \pm \sqrt{j(2n-j)} : 1 \leq j \leq n-1 \right\}. \quad (2)$$

Every listed eigenvalue is simple. The zero eigenspace is spanned by the column spinor whose upper component is $\text{sech}^n x$ and whose lower component is zero.

- (iv) Both $+n$ and $-n$ are resonances and neither is an L^2 eigenvalue.
- (v) Both scalar channels of H_n^2 , and consequently the Dirac scattering problem, are reflectionless.

For $n = 1$, (2) contains only the zero mode; this nontrivial boundary case is included in the theorem.

2 Self-adjoint square and shape invariance

The free off-diagonal derivative operator on H^1 is self-adjoint after the fixed signs in (1) are used. Multiplication by the bounded Hermitian matrix with entries $n \tanh x$ is a bounded self-adjoint perturbation. The Kato–Rellich bounded-perturbation theorem proves assertion (i), and Stone’s theorem supplies the unitary group.

Direct multiplication, using $(\tanh x)' = \text{sech}^2 x$, gives

$$\begin{aligned} B_n &:= A_n^* A_n = -\frac{d^2}{dx^2} + n^2 - n(n+1) \text{sech}^2 x, \\ C_n &:= A_n A_n^* = -\frac{d^2}{dx^2} + n^2 - n(n-1) \text{sech}^2 x, \\ H_n^2 &= \text{diag}(B_n, C_n). \end{aligned} \quad (3)$$

The key identity is the exact shape invariance

$$C_n = B_{n-1} + 2n - 1. \quad (4)$$

Moreover $B_n A_n^* = A_n^* C_n$ and $A_n B_n = C_n A_n$.

Lemma 1 (Scalar hierarchy). *The point spectrum of B_n consists precisely of the simple values*

$$q_{n,j} = j(2n - j), \quad j = 0, \dots, n - 1. \quad (5)$$

Its essential spectrum is $[n^2, \infty)$ and is purely absolutely continuous. At n^2 there is a resonance and no eigenvalue. The same statements for C_n hold with the list $j = 1, \dots, n - 1$.

Proof. The seed $\phi_{m,0} = \text{sech}^m x$ satisfies $A_m \phi_{m,0} = 0$. For $0 \leq j < m$, repeated use of (4) gives the nonzero state

$$\phi_{m,j} = A_m^* A_{m-1}^* \cdots A_{m-j+1}^* \text{sech}^{m-j} x, \quad B_m \phi_{m,j} = j(2m - j) \phi_{m,j}. \quad (6)$$

Its nodal degree, or equivalently the nonvanishing intertwining norms, shows that these states are simple and distinct.

Construction is not yet exhaustion. If a further L^2 eigenstate of B_m had energy below m^2 , applying $A_m, A_{m-1}, \dots$ either lands in one of the one-dimensional kernels already used in (6), or produces an L^2 eigenstate below the continuum of $B_0 = -d^2/dx^2$. The latter has none. Reversing the intertwiners therefore leaves exactly (5).

It remains to prove the spectral type rather than infer it from formal Jost solutions. Put

$$D_m = A_m^* A_{m-1}^* \cdots A_1^*, \quad B_0 = -d^2/dx^2, \quad Q_m = \prod_{r=1}^m (B_0 + r^2).$$

Induction from (4), first on Schwartz functions and then by closure, gives

$$B_m D_m = D_m (B_0 + m^2), \quad D_m^* D_m = Q_m. \quad (7)$$

Since $Q_m \geq (m!)^2$, the initially defined operator $U_m = D_m Q_m^{-1/2}$ extends to an isometry on $L^2(\mathbb{R})$ and satisfies $B_m U_m = U_m (B_0 + m^2)$. Its range is closed and equals the closure of $\text{ran } D_m$; hence its orthogonal complement is $\ker D_m^*$.

Each of the m states in (6) belongs to this kernel: applying the adjoint intertwining would otherwise create an L^2 eigenstate of $B_0 + m^2$ below m^2 . Conversely, D_m^* is an order- m differential operator, so its solution space, and therefore its L^2 kernel, has dimension at most m . Thus

$$(\text{ran } U_m)^\perp = \text{span}\{\phi_{m,0}, \dots, \phi_{m,m-1}\}.$$

The restriction of B_m to the orthogonal complement of these bound states is consequently unitarily equivalent to $B_0 + m^2$. This proves at once that its remaining spectrum is $[m^2, \infty)$, purely absolutely continuous, with no embedded eigenvalue or singular-continuous part. Finally, (4) transfers the complete statement to C_n . $\square$

3 Dirac point spectrum and chirality

The equations $A_n u = 0$ and $A_n^* v = 0$ give $u = c \text{sech}^n x$ and $v = c' \cosh^n x$. Exactly the first is square-integrable, so $\ker H_n$ is one-dimensional and upper chiral. Its squared norm is

$$\int_{\mathbb{R}} \text{sech}^{2n} x \, dx = \frac{4^n n! (n-1)!}{(2n)!}. \quad (8)$$

For every $q > 0$, A_n and A_n^* identify the q -eigenspaces of B_n and C_n . If $B_n u = qu$, the two normalized linear combinations of u and $q^{-1/2} A_n u$ have Dirac energies $\pm \sqrt{q}$. Lemma 1 now proves (2), including all multiplicities. The anticommutation of H_n with $\text{diag}(1, -1)$ is the structural reason for the nonzero sign pairing.